

KRISTIANSAND KJEVIK, Norway

WMO: 014520

Lat: 58.204N

Lon: 8.085E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-10.9	-16.5	0.9	-11.4	-14.0	1.1	-8.2	12.1	5.1	11.1	4.0	3.0	50	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	24.9	17.2	23.2	16.4	21.8	15.8	18.6	22.5	17.7	21.4	16.8	20.2	4.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	19.8	16.2	11.5	19.0	15.3	10.9	18.3	52.6	22.6	49.8	21.5	47.3	20.2	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.9	8.0	-15.9	27.1	3.4	1.8	-18.3	28.4	-20.3	29.5	-22.3	30.5	-24.8	31.9
			-16.2	19.7	3.4	1.1	-18.6	20.5	-20.6	21.1	-22.5	21.7	-24.9	22.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-0.1	0.1	2.2	6.0	10.4	14.1	16.6	15.9	12.6	8.0	3.9	0.6
	DBStd	6.94	4.58	4.31	3.62	3.02	3.12	2.43	2.31	2.31	2.42	3.08	3.63	4.97
	HDD10.0	1554	314	278	243	126	33	1	0	0	6	76	185	292
	HDD18.3	3957	573	511	502	371	248	131	64	81	173	320	434	550
	CDD10.0	659	0	0	1	4	44	123	205	184	83	14	1	0
	CDD18.3	19	0	0	0	0	1	2	11	6	0	0	0	0
	CDH23.3	135	0	0	0	0	2	18	77	37	1	0	0	0
	CDH26.7	8	0	0	0	0	0	2	4	2	0	0	0	0
Wind	WSAvg	3.7	4.0	4.0	3.9	3.7	3.8	3.7	3.5	3.4	3.6	3.6	3.8	3.7
Precipitation	PrecAvg	1382	153	100	78	71	77	83	83	126	136	165	164	138
	PrecMax	1940	410	287	153	180	175	206	221	255	300	317	443	292
	PrecMin	776	25	12	9	11	5	3	2	10	36	55	40	34
	PrecStd	276	82	64	39	41	44	52	52	59	65	72	89	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	11.0	13.9	18.0	23.5	25.5	27.2	26.2	21.2	16.0	13.0	10.5
		MCWB	6.9	7.7	8.8	10.9	15.2	16.0	18.0	17.5	15.5	12.4	10.8	8.7
	2%	DB	7.7	8.1	11.1	15.4	21.0	23.0	25.5	24.1	19.2	14.4	11.6	9.2
		MCWB	6.1	5.8	6.5	9.6	13.8	15.2	17.5	17.3	14.5	12.0	9.8	7.9
	5%	DB	6.8	7.0	9.0	13.2	18.3	21.0	23.8	22.6	18.0	13.7	10.3	8.3
		MCWB	5.4	5.2	5.2	8.2	12.3	14.3	16.9	16.8	14.2	11.7	8.8	7.1
	10%	DB	5.8	5.9	7.2	11.2	16.4	19.4	22.0	20.9	16.9	12.9	9.4	7.2
		MCWB	4.6	4.5	4.5	6.9	11.2	13.6	16.0	16.1	13.8	11.0	8.1	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	8.2	9.1	12.4	16.6	17.7	20.0	19.7	17.5	14.0	11.5	9.1
		MCDB	8.4	10.2	12.6	16.4	21.7	22.4	24.4	22.8	19.3	15.0	12.4	9.9
	2%	WB	6.4	6.6	7.1	10.2	14.7	16.4	18.8	18.6	16.2	13.0	10.1	8.2
		MCDB	7.2	7.5	9.7	14.0	18.9	21.1	23.1	22.1	17.9	14.0	11.2	9.0
	5%	WB	5.5	5.6	6.0	9.0	13.2	15.4	17.8	17.7	15.2	12.4	9.2	7.3
		MCDB	6.4	6.4	7.9	12.3	17.2	19.4	22.0	21.0	16.9	13.4	10.1	8.1
	10%	WB	4.7	4.6	5.1	7.8	11.9	14.5	17.0	16.9	14.4	11.5	8.3	6.1
		MCDB	5.5	5.7	6.8	10.2	15.4	17.9	20.6	19.8	16.2	12.7	9.2	7.1
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.0	7.1	8.0	8.5	8.2	8.3	8.1	7.2	6.5	5.6	5.5
		MCDBR	6.2	6.8	10.0	11.2	12.0	11.2	11.7	10.6	8.9	7.1	6.4	5.7
		MCWBR	5.6	5.1	6.2	6.4	6.3	5.9	5.6	5.4	5.5	5.3	5.8	5.3
	5% WB	MCDBR	5.6	5.8	8.2	9.4	10.4	9.6	9.8	8.8	7.1	5.6	5.8	5.6
MCWBR		5.2	4.9	5.7	5.8	6.0	5.7	5.4	5.2	5.2	5.0	5.6	5.5	
Clear-Sky Solar Irradiance	taub	0.293	0.321	0.339	0.384	0.374	0.360	0.379	0.380	0.357	0.334	0.304	0.271	
	taud	2.244	2.326	2.370	2.300	2.404	2.466	2.430	2.423	2.479	2.459	2.313	2.193	
	Ebn at Noon	543	711	809	822	860	878	850	821	785	695	530	444	
	Edh at Noon	47	70	87	110	106	101	103	97	79	61	44	36	
All-Sky Solar Radiation	RadAvg	0.34	0.92	2.31	3.88	5.23	5.90	5.36	4.12	2.67	1.25	0.46	0.24	
	RadStd	0.07	0.19	0.23	0.48	0.59	0.47	0.54	0.29	0.29	0.19	0.10	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-120	-161	N/A	N/A

Nomenclature: See separate page